

International Workshop on Engineering Explainability for Intelligent Software Systems (ExISSt)

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Abstract—In our modern society, software systems are highly integrated into our daily life, control critical infrastructure and advance innovation across multiple domains such as transportation, healthcare and robotics. With the increasing complexity of tasks and higher degrees of system autonomy, algorithms have become more complex and opaque in their decisions. Explainability has been identified as a means to achieve transparency. While focusing on explainable Artificial Intelligence (XAI) in the beginning, communities from other areas of knowledge (e.g., human-computer interaction, philosophy, psychology, and cyber-physical systems) have also started researching explainability over the last years. Explainability needs to be carefully considered throughout the whole software engineering process. From requirements elicitation and design to implementation, testing, and maintenance. Consequently, recent work has broadened its perspective, tackling topics such as requirements engineering for explainability, quality models, and methods for designing and evaluating explanation. However, these efforts remain fragmented across different communities. This workshop aims to provide a dedicated forum to consolidate these efforts within the software engineering community. By bringing together researchers and practitioners from the different fields involved, the workshop fosters the exchange of ideas, strengthens links between communities, and works towards establishing explainability throughout the software engineering lifecycle.

Index Terms—explainability, transparency, trustworthy systems, intelligent software systems

I. ORGANIZERS

A. Organizing Committee

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- Ali Torbati, Carl von Ossietzky Universität Oldenburg, Germany (**proceedings co-chair**)
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B. Steering Committee

- Eric Knauss, Chalmers|University of Gothenburg, Sweden

- Kurt Schneider, Leibniz University of Hannover, Germany
- Mersedeh Sadeghi, University of Cologne, Germany
- Andreas Vogelsang, University of Duisburg-Essen, Germany
- Maike Schwammberger, Karlsruhe Institute of Technology, Germany
- Timo Speith, University of Bayreuth, Germany

II. MOTIVATION AND OBJECTIVES

A. Motivation

The growing autonomy and complexity of intelligent software systems have made their decisions increasingly opaque, not least due to the progress and increasing popularity of artificial intelligence (AI) approaches. This opacity is problematic because such systems now operate in domains such as transportation, healthcare, and robotics, where incomprehensible behavior can undermine safe operation, regulatory compliance, and user acceptance.

Explainability addresses this problem on several levels. As a software quality aspect, it has been identified as a means to achieve ethics, fairness, and transparency. These properties are widely regarded as essential for trustworthy (AI-based) systems [2]. Explainability improves human-computer interaction by providing a level of understandability required to use these technologies efficiently and safely, and it fosters users' trust in systems [3]. Additionally, With regulatory frameworks such as the EU AI Act introducing concrete transparency obligations, explainability is moreover evolving from a desirable property into a legal requirement [11], [15].

Crucially, explainability must be addressed as a core design requirement. Like other non-functional properties, it must be systematically addressed throughout the entire software engineering lifecycle [4]. However, research on such systematic explainability engineering is still scarce, and the first RE4ES workshop identified a long list of open research questions spanning multiple software engineering disciplines [1].

While focusing on explainable AI (XAI) in the beginning, communities from other research disciplines (e.g., philosophy, psychology, law and human-computer interaction) have started

researching broader conceptions of explainability over the last years [10]. A growing interest in this topic can also be seen in the software engineering community as indicated by the growing number of papers on explainability at co-located events such as SEAMS [7], CAIN [14], CHASE [8], SEIP [13], LLM4Code [12] and JAWs [16]; or the newly established Track on Explainable and Trustworthy Applications at the Euromicro Conference on Software Engineering and Advanced Applications (SEAA) conference in 2026.

Given the pressing practical and regulatory needs, the lack of systematic research, and a growing but still fragmented community, we believe that a dedicated forum at ICSE is both timely and necessary.

B. Objectives

The workshop has three objectives: advancing research on systematic engineering for explainable systems and software, community building, and interdisciplinary exchange. We aim to identify and tackle open research problems in the area of explainability together with the research community of ICSE, similar to the research roadmap in [1] that provides challenges for different disciplines of software engineering with explainability as a first-class citizen:

- Defining and Measuring Explainability
- Considering Stakeholders and Contexts
- Considering Goals and Desiderata for Explanations
- Creating Explanations: Models, Languages, Design, Algorithms
- Design Processes
- Quality Aspects and Testing
- Quantifying and Reducing Costs and Resources
- AI-Based Software Engineering
- Explainable AI generated Software Codes
- Explainability for Developers vs Users
- Effect of EU AI Act on Software Engineering

Due to the interdisciplinary nature of the topic, we place special emphasis on community building through group activities, discussion sessions, and a social event. These activities will strengthen and expand the research community by promoting collaboration among researchers from diverse backgrounds, facilitating networking, and establishing connections that can lead to future joint initiatives.

C. Additional outcomes

Besides the proceedings, a specific outcome of the workshop will be a research roadmap identifying and prioritizing the key open challenges of engineering explainability. This roadmap will consolidate the perspectives of the participating communities into a shared research agenda, providing orientation for future work in the field. We plan to develop the roadmap collaboratively during dedicated discussion sessions and to publish it as a workshop report, making it available to the broader software engineering community.

III. FORMAT AND REQUIRED SERVICES

A. Workshop Format

The workshop will be a half a day up-to a full day workshop, including a mixture of invited talks and paper presentations depending on the number of submission, and a break-out session to facilitate collaboration and discussions. We conclude the day with a social event (dinner) for participants having more time to not only get to know each other better but also brainstorming their ideas on the topic. To enrich the discussions of the paper presentations, we ask authors to read one other paper in advance and to prepare questions for the discussion round.

We will hold the talks and paper presentation in 2 consecutive sessions in order to facilitate enough time for a lively discussion. To ensure that the debate is dynamic, we will ask the authors of another paper in the session to prepare at least one critical question for other authors. These questions will be asked at the end of each presentation. Furthermore, we will collect ideas and wishes for further workshops and community-building events.

B. Services

No special services are required: standard services (room, technical equipment) will suffice. A preliminary list of registered participants would be of help for planning dissemination activities.

IV. TARGET AUDIENCE

The workshop is directed at researchers from any area who work on or are interested in the topic of explainability, as well as practitioners who already have experience in dealing with explainability in practice or who face the challenge of making their systems explainable. Given the interdisciplinary nature of the topic and the workshop's community-building goals, we explicitly welcome participants from all involved disciplines and at all career stages.

Our workshop format is designed for a smaller, focused audience. Previous editions at the Requirements Engineering Conference (2021-2023) attracted around 20-35 participants. In 2024, we moved the workshop to ICSE to reach the broader software engineering community; this first edition at a new venue drew a smaller audience, as AI was still an emerging topic. With the rise of AI in software engineering, explainability is becoming more important so that we expect approximately 30 participants this year. The workshop is open to anyone interested in the topic who would like to engage with and contribute to a growing research community.

Solicitation and Selection of Participants. The workshop will be open to all interested researchers and practitioners. We will solicit participation through a call for papers distributed via the relevant mailing lists of the software engineering and neighboring communities (e.g., SEWORLD, and the mailing lists of related conferences and special interest groups), as well as through social media and the workshop website. In addition, we will directly invite a curated list of authors and participants from previous editions of our workshop.

V. PROCEEDINGS

Workshop proceedings will be published with IEEE, the standard option offered to ICSE workshops.

A. Contribution types

The workshop solicits two contribution types:

- Research papers (max. 8 pages with references). These papers should describe ongoing research or preliminary results aligned with the workshop's topics of interest. They should already have a proof of concept or a distinct vision of what the research aims at. They could be improvements of existing approaches, as well as empirical studies, and experience reports (e.g., of applied work in the industry).
- Short papers (max. 5 pages with references). These papers serve to foster discussion on hot, relevant topics in the field. They could outline rough, but plausible visions regarding the future of explainability, as well as problem statements explaining problems in industrial settings.

In both cases, papers will be peer-reviewed and accepted papers will be published in the workshop proceedings. Submissions should clearly describe how they relate to the workshop scope and objectives.

B. Review and Evaluation Process

The selection of papers will be based on workshop relevance, academic rigour and sound argumentation, innovation, practical relevance, and quality of writing. Position/vision papers will further be evaluated on their potential for generating discussions and the originality of the vision for the field. Every paper will be peer-reviewed by at least two members of the program committee adopting a single-blind method by default. Submission, reviewing, and communication will be supported through EasyChair and will be managed by our proceeding chair.

C. Program Committee

Our program committee consists of a mix of active researchers, who have previously published in this field or the relevant fields. We focus on a mix of academics, including from an industry background or who have worked on industry projects. We also plan to invite all PC members from last year. The current list of potential PC members is:

- Anastasia Mavridou, NASA Ames Research Center, USA
- Rasmus Adler, Fraunhofer IESE, Germany
- Douglas Cirqueira, Dublin City University, Ireland
- Luiz Marcio Cysneiros, York University, Canada
- Joel Greenyer, FHDW Hannover, Germany
- Holger Hermanns, Saarland University, Germany
- Marjo Kauppinen, Aalto University, Finland
- Marie Farell, University of Manchester, UK
- Yasmeen Rafiq, University of Manchester, UK
- Genaina Rodrigues, University of Brasília, Brasil
- Livia Lestingi, Politecnico di Milano
- Ricardo Caldas, Gran Sasso Science Institute, Italy
- Jil Klünder, FHDW Hannover, Germany

- Matthias Koch, Fraunhofer IESE, Germany
- Marco Konersmann, Fern University Hagen, Germany
- Felix Lindner, University of Ulm, Germany
- Tim Miller, University of Melbourne, Australia
- Sven Peldszus, Chalmers|University of Gothenburg, Sweden
- Jie Zou, University of York, UK
- Simos Gerasimou, University of York, UK

as well as the members from our steering committee. Proceedings Chair and the steering committee are listed in Section I-A.

D. Workshop Website

The website for our workshop will be available at <https://gricel-lee.github.io/ExISSt-ICSE27/> (URL will be updated after acceptance). All necessary information such as key dates for our workshop will be published there.

VI. WORKSHOP HISTORY

Although we are proposing this workshop as a newly established event rather than as a direct successor to previous workshops, our team has substantial experience in organizing explainability-related workshops in different communities and international venues.

The first edition of the RE4ES workshop was co-located with the 29th IEEE International Requirements Engineering Conference (RE 2021) in Notre Dame, South Bend, USA. It was held online on September 20 and 21, 2021, with around 20 participants.

The second edition was co-located with the 30th IEEE International Requirements Engineering Conference (RE 2022) in Melbourne, Victoria, Australia. It was held as a hybrid event on August 15, 2022, consisting of online participants and a local hub in Oldenburg, Germany. We had around 35 participants.

The third edition was co-located with the 31st IEEE International Requirements Engineering Conference (RE 2023) in Hannover, Lower Saxony, Germany, with around 30 participants.

In 2024, The fourth edition was co-located with the 46th ACM/IEEE International Conference on Software Engineering (ICSE 2024) in Lisbon, Portugal to reach the broader software engineering community; while this first edition at a new venue drew a smaller audience (around 10 participants), it laid the groundwork for establishing the topic at the premier software engineering conference.

VII. ORGANIZERS' BIOS

Verena Klös is a Professor for Embedded Hardware-/Software Systems Chair at Carl von Ossietzky Universität Oldenburg since 2024. Before, she was a junior professor of Tactile Computing at Technical University of Dresden in the Centre for Tactile Internet with Human-in-the-Loop (CeTI). She received her PhD from Technical University Berlin in 2020. In her phd thesis, she developed a framework for safe, intelligent and explainable self-adaptive systems. Currently, her research focuses on explainability of autonomous and

cyber-physical systems. She has co-organized the RE4ES workshop (2021-2023) and the special session on cyber-physical systems at the Euromicro Conference on Software Engineering and Advanced Applications (SEAA) since 2021. In 2026 she is PC Co-Chair of the SEAA conference.

Gricel Vázquez is a Research Fellow at the University of York, UK, where she completed her PhD studies in Computer Science. She received her MSc in Computational Intelligence and Robotics at the University of Sheffield with distinction. Her research focus on formal methods, verification of AI-enabled intelligent systems, self-adaptive systems, explainability for task planning problems and model-driven engineering. She has been part of the organising committee of multiple conferences and workshops including SEAMS@ICSE'27, FMS'26, REXAI@NFM'26, ICPE'25, ACSOS'25, FAACS@ICSA'25, FMS'24-'23, and SEAMS@ICSE'23.

Ali Torbati is a Research Associate (PhD Candidate) at the Carl von Ossietzky Universität Oldenburg. Shortly before, he was a research associate at Technical University of Dresden in the 6G-life Project. He received his MSc in Computer Engineering at the Eastern Mediterranean University (Cyprus) in 2023. His research focus on self-adaptive systems, explainability and optimization.

ACKNOWLEDGMENT

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VIII. DRAFT CALL FOR SUBMISSIONS

In our modern society, software systems are highly integrated into our daily life, control important processes in industry and transportation, and advance medical treatments. With the increasing complexity of tasks and higher degrees of system autonomy, algorithms have become more complex and opaque in their decisions. Thus, quality aspects such as ethics, fairness, and transparency have become essential for trustworthy software systems. Explainability has been identified as a means to achieve these aspects and has become a hot topic. While focusing on explainable Artificial Intelligence (XAI) in the beginning, communities from other areas of knowledge (e.g., human-computer interaction, philosophy, psychology, and cyber-physical systems) have also started researching explainability in the last years.

Explainability needs to be carefully considered during the whole software engineering process. However, there is only little research on systematic explainability engineering yet but a long list of open research questions with challenges for different disciplines of software engineering [1].

We aim to tackle these open research problems. Due to the interdisciplinary nature of the topic, we also focus on community building to strengthen links in the community and foster communication between researchers working in the different fields.

A. Topics of Interest

The topics of interest include but are not limited to:

- Explanation visualization and interfaces
- Elicitation and analysis techniques for explainable systems
- Specification and validation of explainability requirements
- Perspectives and views of different stakeholders
- Impact of explainability on other software quality aspects; and dealing with antagonistic quality aspects
- Modeling and specification languages
- Formal validation and verification techniques for explainable systems
- Architectures for explainable systems
- Challenges for explainability in practice

B. Contribution Types

The workshop solicits two contribution types:

- Research papers (max. 8 pages with references). These papers should describe ongoing research or preliminary results aligned with the workshop's topics of interest. They should already have a proof of concept or a distinct vision of what the research aims at. They could be improvements of existing approaches, as well as empirical studies, and experience reports (e.g., of applied work in the industry).
- Short papers (max. 5 pages with references). These papers serve to foster discussion on hot, relevant topics in the field. They could outline rough, but plausible visions

regarding the future of explainability, as well as problem statements explaining problems in industrial settings.

C. Important Dates

- Deadline for submission: Thu 27 Nov 2026
- Notification of accepted papers: Fri 11 Dec 2026
- Camera-ready paper due: Fri 29 Jan 2027
- Workshop Day: between 25 Apr - 1 May in Dublin, Ireland.

D. Submission Guidelines

All submissions must conform to the IEEE conference proceedings template, specified in the IEEE Conference Proceedings Formatting Guidelines (title in 24pt font and full text in 10pt type, LaTeX users must use `\documentclass[10pt,conference]{IEEEtran}` without including the `compsoc` or `compsocconf` options). Note that IEEE format is being used this year, whereas last year it was ACM format.

- All submissions must not exceed the respective page limit **including** all figures, tables and appendices.
- Submissions must strictly conform to the IEEE Conference Proceedings Formatting Guidelines specified above. Alterations of spacing, font size, and other changes that deviate from the instructions may result in desk rejection without further review.
- By submitting to the workshop, authors acknowledge that they are aware of and agree to be bound by the ACM Policy and Procedures on Plagiarism and the IEEE Plagiarism FAQ, that they conform to the authorship policy of the ACM, and the authorship policy of the IEEE, and that they acknowledge that they are subject to all ACM Publications Policies, including ACM's new Publications Policy on Research Involving Human Participants and Subjects. Alleged violations of this policy or any ACM Publications Policy will be investigated by ACM and may result in a full retraction of your paper, in addition to other potential penalties, as per ACM Publications Policy.
- Please ensure that you and your co-authors obtain an ORCID ID, so you can complete the publishing process for your accepted paper.

E. Submission URL

Submit via EasyChair: link tba

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- [1] Brunotte, Wasja, Larissa Chazette, Verena Klös, and Timo Speith. "Quo vadis, explainability?—a research roadmap for explainability engineering." In international working conference on requirements engineering: foundation for software quality, pp. 26-32. Cham: Springer International Publishing, 2022.